

DATA 366: Introduction to Multivariate Statistics

Course Description: This course covers statistical methods used for analysis of data sets that contain more than one variable. Topics include multivariate normal distribution, covariance, inference about a mean vector, MANOVA, regression, and classification.

Course prerequisites: Linear Algebra (MATH 313 or MATH 310 or MATH 215 or DATA/MATH 412) and Introduction to Statistical Methods (DATA/MATH 363).

Instructor and Contact Information

Professor/Lecturer: Dr. Nazish Shahid

Office: TBA

Phone:

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Class meetings: Sections A, B: Tuesday, Friday (8:00-9:40 AM); Shensi Building 404 Room

Sections C, D: Tuesday (9:55-11:35 AM); Shensi Building 414 Room

Sections C, D: Friday (9:55-11:35 AM); Shensi Building 120 Room

Office hours: TBA

Textbooks: Applied Multivariate Statistical Analysis (6th Edition), Pearson by Richard A. Johnson, Dean W. Wichern

Course Objectives: Multivariate Statistical techniques are important tools of analysis in all domain sciences. During this course students will:

- Have the knowledge of the basic concepts underlying the most important multivariate techniques;
- Address both the underlying mathematics and problems of applications;
- Select appropriate methods of multivariate data analysis, given multivariate data and study objectives.

Learning Outcomes: Students will be able to:

- Identify different multivariate analysis methods' strengths and weaknesses;
- Implement the statistical methods and interpret results of multivariate data analyses;
- Write R programs to carry out multivariate data analyses, develop and troubleshoot R codes for running statistical tests and generating graphics.

Major topics:

Weeks	Topics
Week 1	Aspects of Multivariate Analysis
Week 2- Week 3	Matrix Algebra and Random Vectors
Week 4 - Week 5	Multivariate Normal Distribution
Week 6 – Week 7	Inference about a Mean Vector, Midterm Exam
Week 8 – Week 10	Analysis of Covariance Structure
Week 11- Week 13	Classification and Discrimination, Midterm Exam
Week 14 – Week 15	Clustering
Week 16	Review and Final exam preparation

Homework: Weekly homework will be assigned.

Exams: There will be two in-class midterm tests. If you have to miss the exam for a university approved reason, your final exam score (percentage) will be used as the score for that exam.

Final Exam: There will be a comprehensive final examination, given according to the University final exam schedule. Please note the following:

- University rules relating to final examinations may be found at:

<https://www.registrar.arizona.edu/courses/final-examination-regulations-and-information>

- The University final exam schedule may be found at:

<http://www.registrar.arizona.edu/schedules/finals.htm>

Course grade: Your grade will be determined by homework (20%), midterm tests (25% each), final test (30%). Grades will be no lower than the following:

A: 100-90% B: 89-80% C: 79-70% D: 69-60% E: 59-0%

Requests for incomplete (I) or withdrawal (W) must be made in accordance with university policies, which are available at

<http://catalog.arizona.edu/policy/grades-and-grading-system#incomplete> and
<http://catalog.arizona.edu/policy/grades-and-grading-system#Withdrawal>

respectively.

A student may withdraw from the course with a deletion from record through the posted UAccess date (include official date). A student may withdraw with a grade of "W" through (include official date), using UAccess. It is suggested that students consult his/her academic advisor before withdrawal from any course.

Communication with Students

Announcements and important course information may be sent out via official University email or through D2L. It is the student's responsibility to check for messages and announcements regularly.

Accessibility and Accommodations

At the University of Arizona, we strive to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability or pregnancy, you are welcome to let me know so that we can discuss options. You are also encouraged to contact Disability Resources (520-621-3268) to explore reasonable accommodation.

If our class meets at a campus location: Please be aware that the accessible table and chairs in this room should remain available for students who find that standard classroom seating is not usable.

Attendance/Administrative Drops

Daily attendance is expected from every student. Students who miss the first two class meetings will be administratively dropped unless they have made other arrangements. In addition, students with more than 3 unexcused absences may be administratively dropped from the course. (The UA's policy concerning Class Attendance, Participation, and Administrative Drops is available at:

<http://catalog.arizona.edu/policy/class-attendance-participation-and-administrative-drop>).

- The UA policy regarding absences for any sincerely held religious belief, observance or practice will be accommodated where reasonable,

<http://policy.arizona.edu/human-resources/religious-accommodation-policy>.

- Absences pre-approved by the UA Dean of Students (or Dean's designee) will be honored.

See: <https://deanofstudents.arizona.edu/absences>

It is the student's responsibility to notify the instructor in advance of an absence related to religious observation or an activity for which a Dean's excuse has been granted, and to arrange for how any missed work will be handled.

Code of Academic Integrity

Students are encouraged to share intellectual views and discuss freely the principles and applications of course materials. However, graded work/exercises must be the product of independent effort unless otherwise instructed. Students are expected to adhere to the UA Code of Academic Integrity as described in the UA General Catalog.

See: <http://deanofstudents.arizona.edu/academic-integrity/students/academic-integrity>.

UA Nondiscrimination and Anti-harassment Policy

The University is committed to creating and maintaining an environment free of discrimination; see <http://policy.arizona.edu/human-resources/nondiscrimination-and-anti-harassment-policy>

Classroom Behavior Policy

To foster a positive learning environment, students and instructors have a shared responsibility. We want a safe, welcoming, and inclusive environment where all of us feel comfortable with each other and where we can challenge ourselves to succeed. To that end, our focus is on the tasks at hand and not on extraneous activities (e.g., texting, chatting, reading a newspaper, making phone calls, web surfing, etc.).

Students are asked to refrain from disruptive conversations with people sitting around them during lecture. Students observed engaging in disruptive activity will be asked to cease this behavior. Those who continue to disrupt the class will be asked to leave lecture or discussion and may be reported to the Dean of Students.

Some learning styles are best served by using personal electronics, such as laptops and iPads. These devices can be distracting to other learners. Therefore, students who prefer to use electronic devices for notetaking during lecture should use one side of the classroom.

Threatening Behavior Policy

The UA Threatening Behavior by Students Policy prohibits threats of physical harm to any member of the University community, including to oneself.

See <http://policy.arizona.edu/education-and-student-affairs/threatening-behavior-students>.

Subject to Change Statement

Information contained in the course syllabus, other than the grade and absence policy, may be subject to change with advance notice, as deemed appropriate by the instructor.